Requirements Table

Smart Parking Space Rental & License Validator

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow a Vehicle Owner to view currently available parking bays and reserve an available bay for a selected parking period.	High	**Pass:** The selected available bay is successfully reserved and is no longer shown as available for the reserved period. **Fail:** The same bay can be simultaneously reserved by multiple users.	Real-time reservation prevents parking conflicts and improves parking-space utilization.
FR-002	Functional	The system shall capture the vehicle license plate at the parking entrance and validate it against an active reservation before permitting entry.	High	**Pass:** A vehicle with a valid matching reservation is successfully verified. **Fail:** A vehicle without a valid matching reservation is approved.	License-plate validation ensures that only authorized vehicles use reserved parking spaces.
FR-003	Functional	The system shall record the vehicle entry time when a valid vehicle enters the parking facility and record the exit time when the vehicle leaves.	High	**Pass:** Correct entry and exit timestamps are stored for the parking session. **Fail:** Entry or exit timestamps are missing or incorrectly recorded.	Accurate timestamps are required for session tracking, billing, and overstay detection.
FR-004	Functional	The system shall calculate the total parking fee upon vehicle exit based on the elapsed parking duration and the applicable progressive hourly pricing tier.	High	**Pass:** The calculated amount correctly corresponds to the recorded parking duration and applicable pricing tier. **Fail:** An incorrect duration or pricing tier is used.	Accurate progressive billing ensures that Vehicle Owners are charged correctly.
FR-005	Functional	The system shall monitor active parking sessions and flag vehicles that remain parked beyond their authorized reservation period for review by the Parking Warden.	High	**Pass:** A vehicle exceeding its authorized parking period is flagged for the Parking Warden. **Fail:** An unauthorized overstay is not detected or flagged.	Automatic overstay detection allows Parking Wardens to handle unauthorized parking efficiently.
NFR-001	Performance & Security	The gate-camera license-plate lookup and reservation-validation operation shall complete within 300 milliseconds under normal and simulated peak-load conditions.	High	**Pass:** Validation completes within 300 ms while maintaining required security checks. **Fail:** Validation exceeds 300 ms or bypasses security checks.	Fast and secure validation prevents delays and unauthorized entry at the parking gate.
NFR-002	Reliability	The system shall preserve confirmed parking reservations and recover normal reservation-validation functionality within 60 seconds after a temporary service interruption.	Medium	**Pass:** Confirmed reservations remain intact and validation functionality is restored within 60 seconds. **Fail:** Reservations are lost or recovery exceeds 60 seconds.	Reliable recovery minimizes parking-service disruption and prevents loss of reservation information.